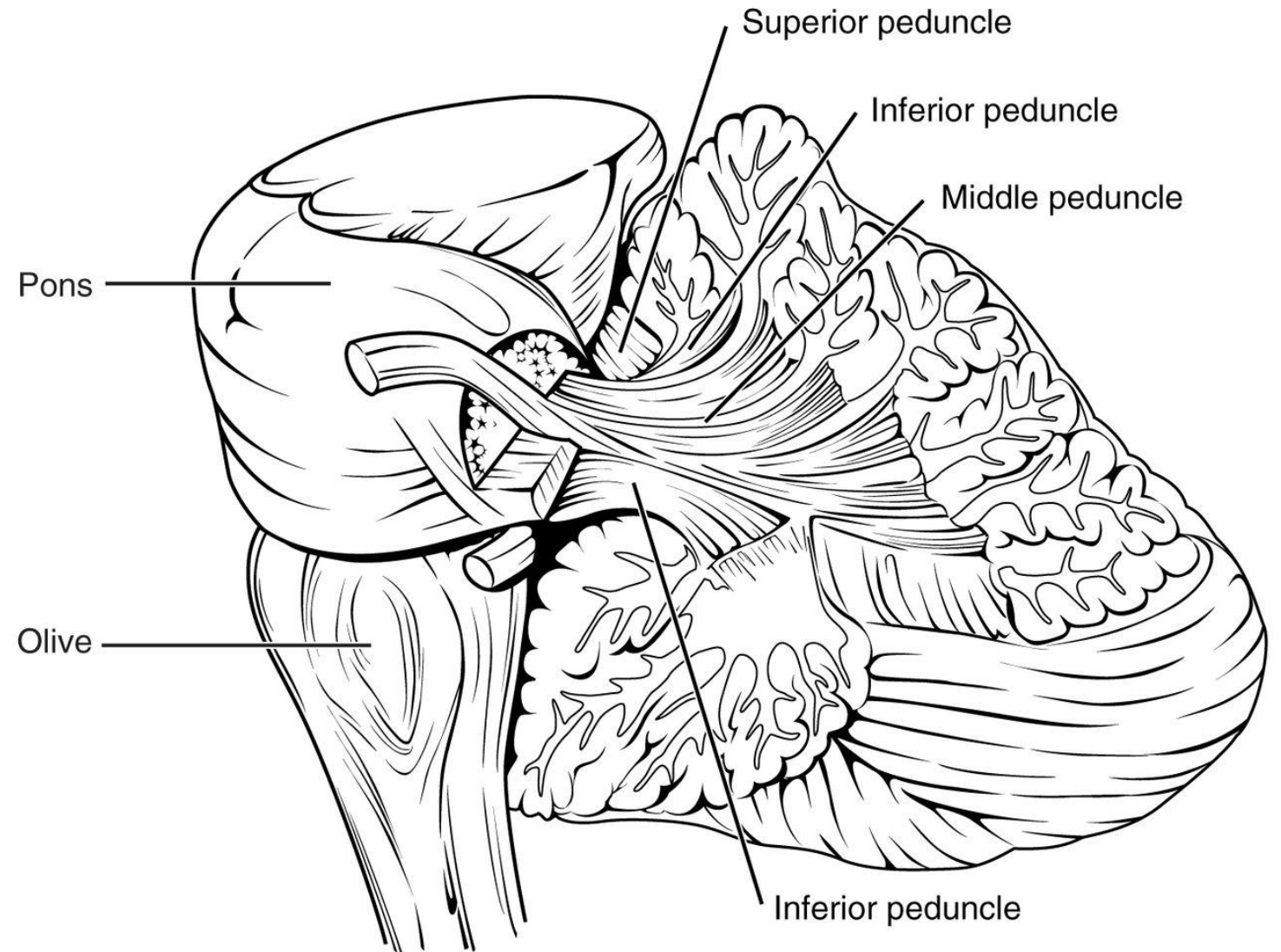


Cerebellum And Cerebellar Deficits



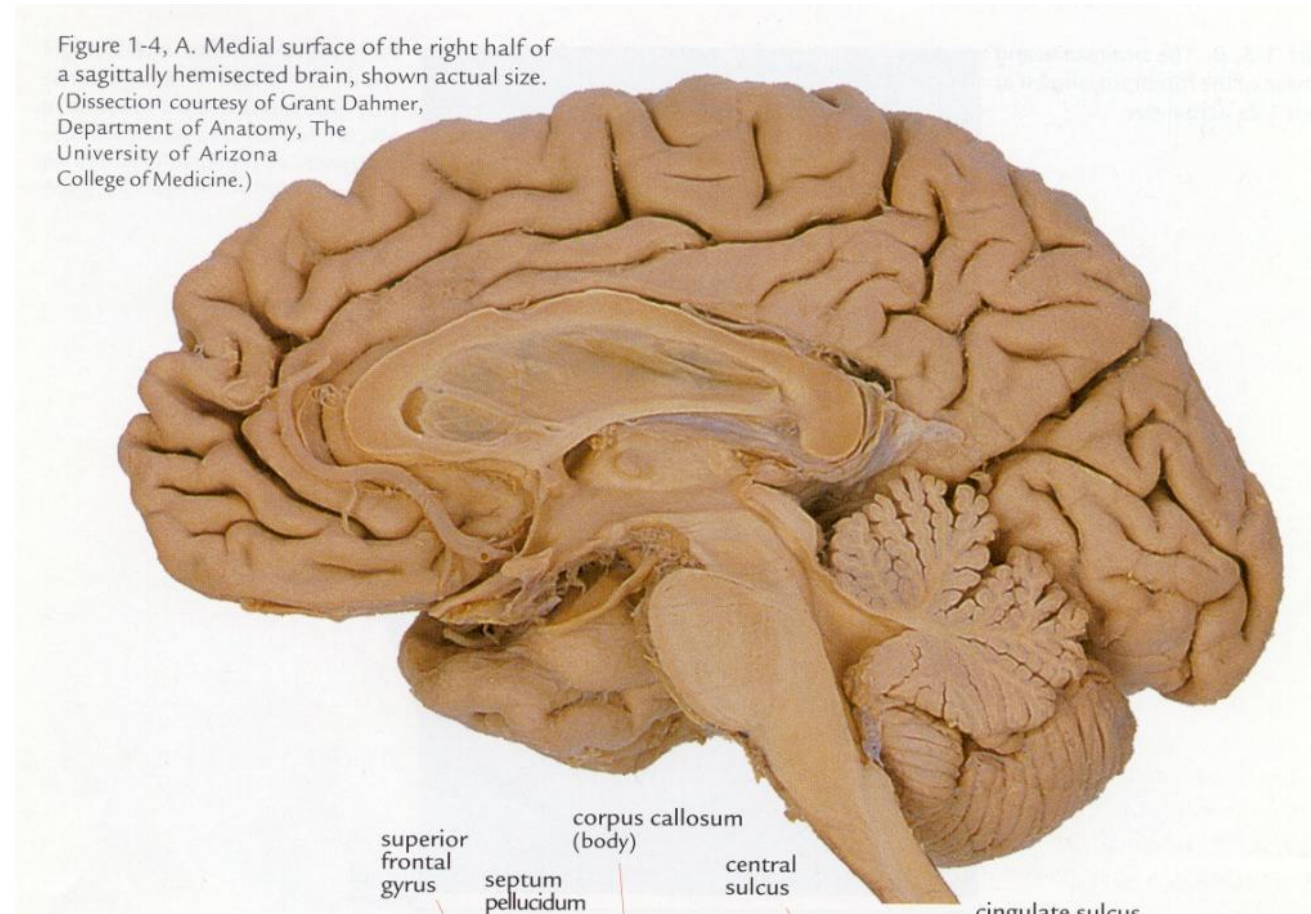
Learning Objectives

1. Explain the overall function of the cerebellum and the role it plays in motor control.
2. Describe the cerebellar anatomy: surfaces, fissures and lobes.
3. Explain the histological organization of the cerebellar cortex and name the cells found in each layer.
4. Describe the functional cerebellar regions.
5. Indicate the origin and name of the afferent pathways providing inputs to the cerebellum.
6. Explain the cerebellar processing.
7. Indicate the important efferent or output fibers passing through each of the cerebellar peduncles and their function.
8. Describe the deficits associated with damage to the cerebellum or the cerebellar peduncles.
9. Name the blood vessels that supply the cerebellum and explain the deficits that result from their occlusion.
10. Briefly explain other acute/chronic cerebellar disorders

OBJ. # 1

Cerebellar Functions

- Control of ongoing movement
 - Contribute to motor planning
 - Detecting motor error
 - Correct the error during movement
 - Stores the correction as memory
- Controls posture and gait and eye mvmts
- Helps to regulate muscle tone



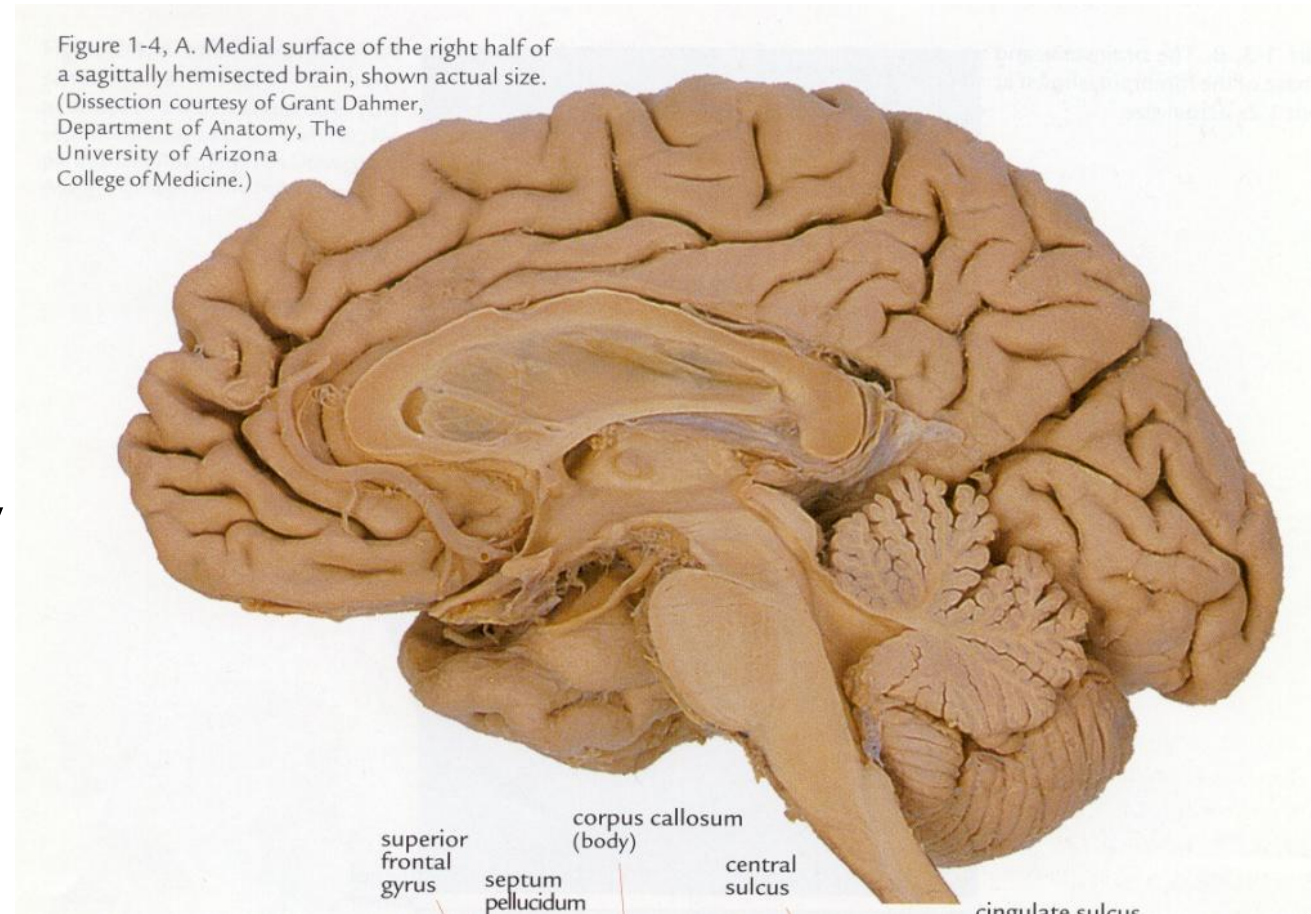
activate agonist and inactivate antagonist muscles

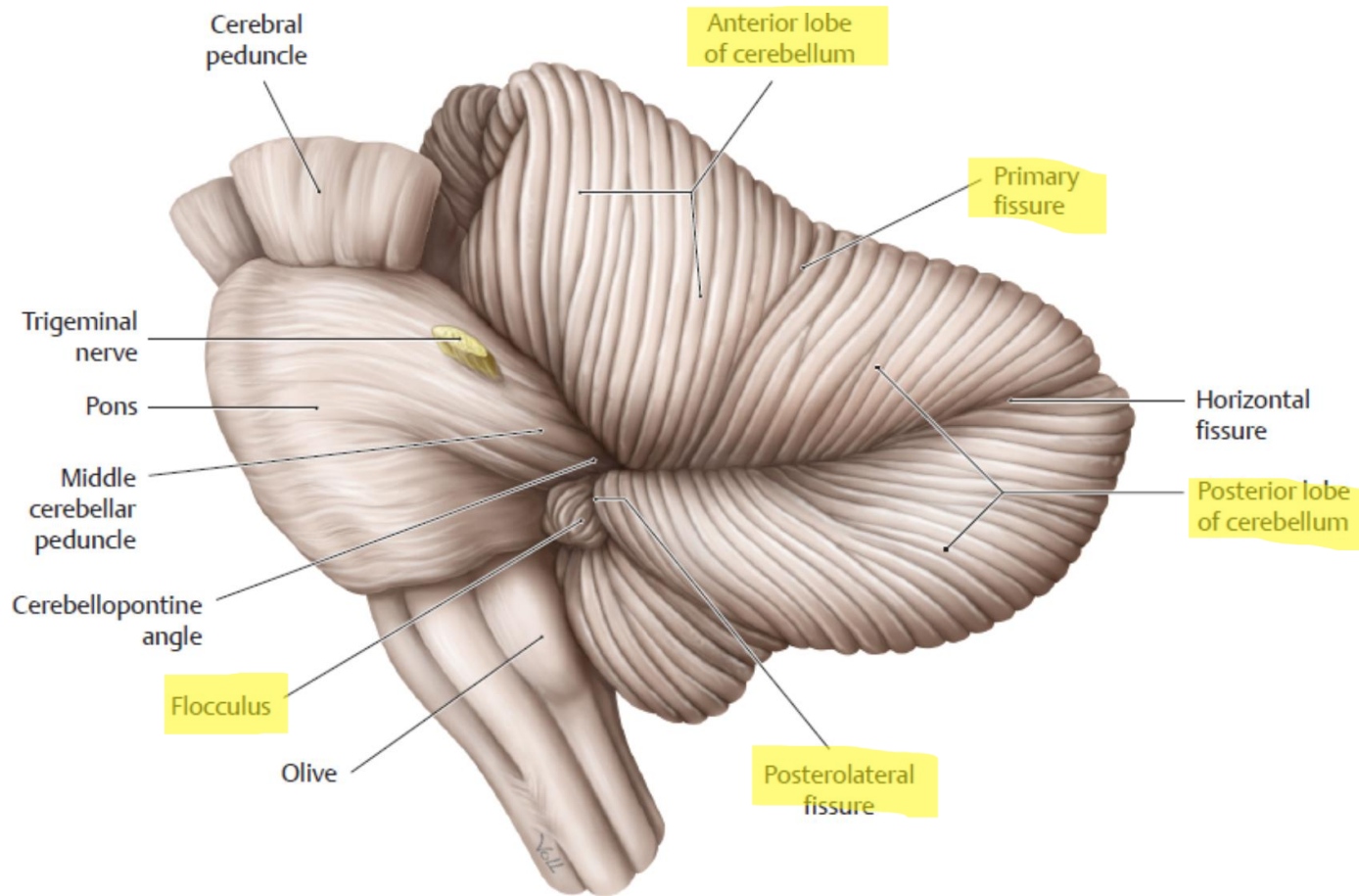


OBJ. # 1

Cerebellar Functions

- Monitor and modulate motor activity originating in other structures.
- Ex: At the end of a movement, automatic excitation of antagonist muscles and inhibition of agonist muscles - allows limbs to arrive at a precise location in space.



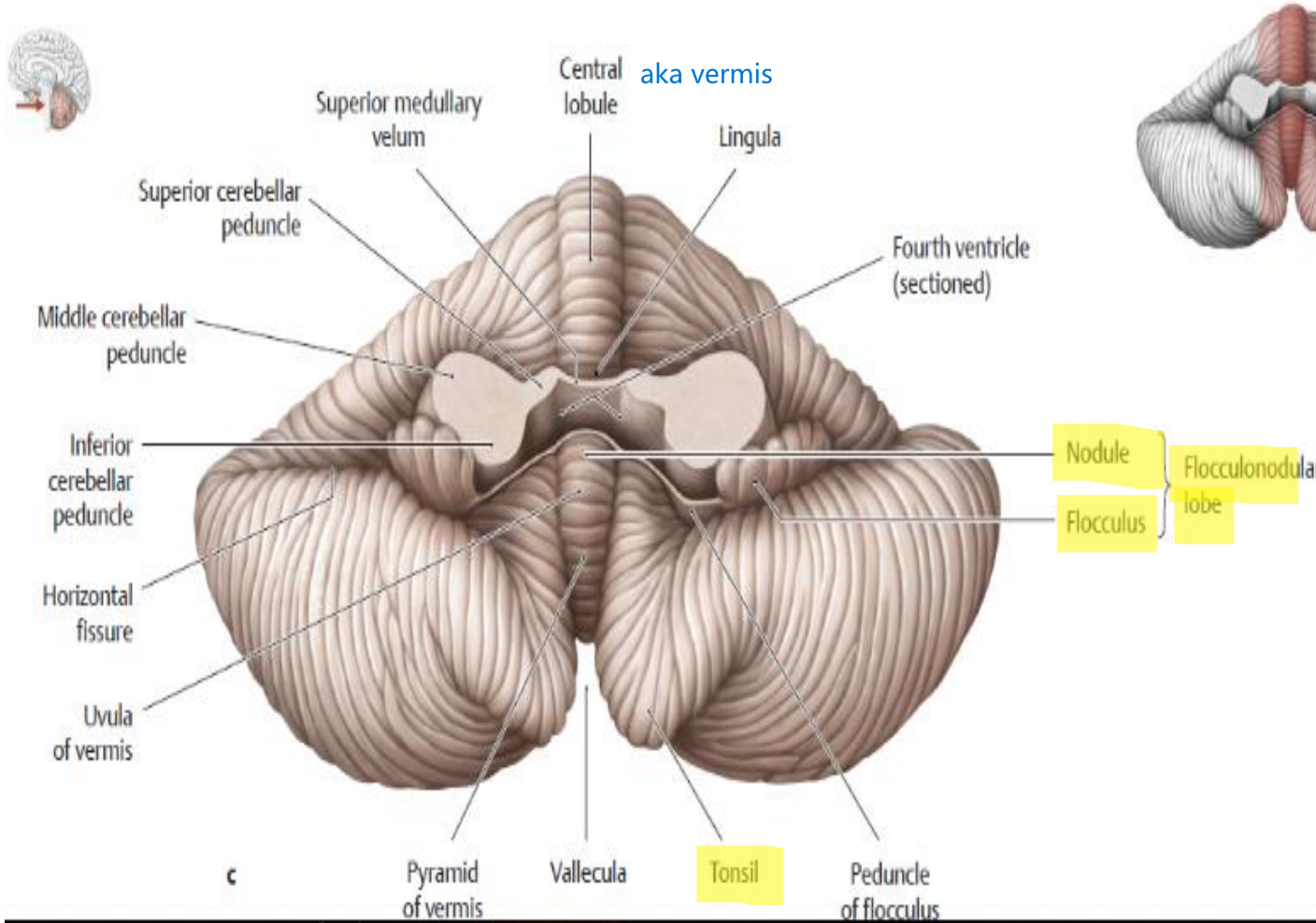


Cerebellar Anatomy

- **3 lobes:** **anterior**, **posterior** and **flocculonodular**
- **2 fissures:** **primary** and **posterolateral**
- **3 surfaces:** **superior**, **inferior**, and **ventral**

★ There is a horizontal fissure, however it has no clear functional or clinical significance

OBJ. # 2



Cerebellar Anatomy

- **3 lobes:** **anterior**, **posterior** and **flocculonodular**
- **2 fissures:** **primary** and **posterolateral**
- **3 surfaces:** **superior**, **inferior**, and **ventral**

★ There is a horizontal fissure, however it has no clear functional or clinical significance

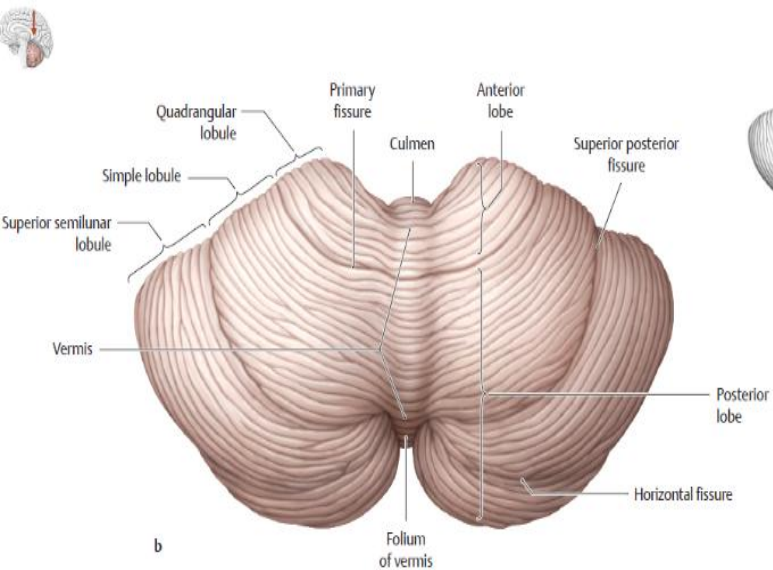
tonsil herniate through foramen magnum and crowd medulla

OBJ. # 2

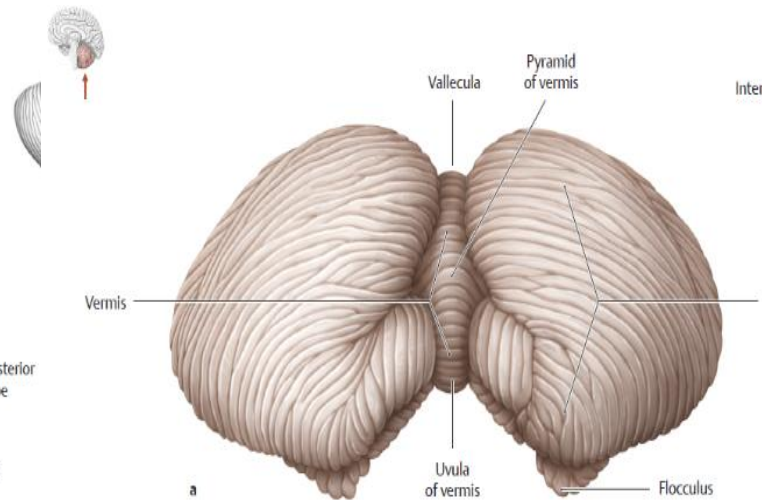
Cerebellar Anatomy

OBJ. # 2

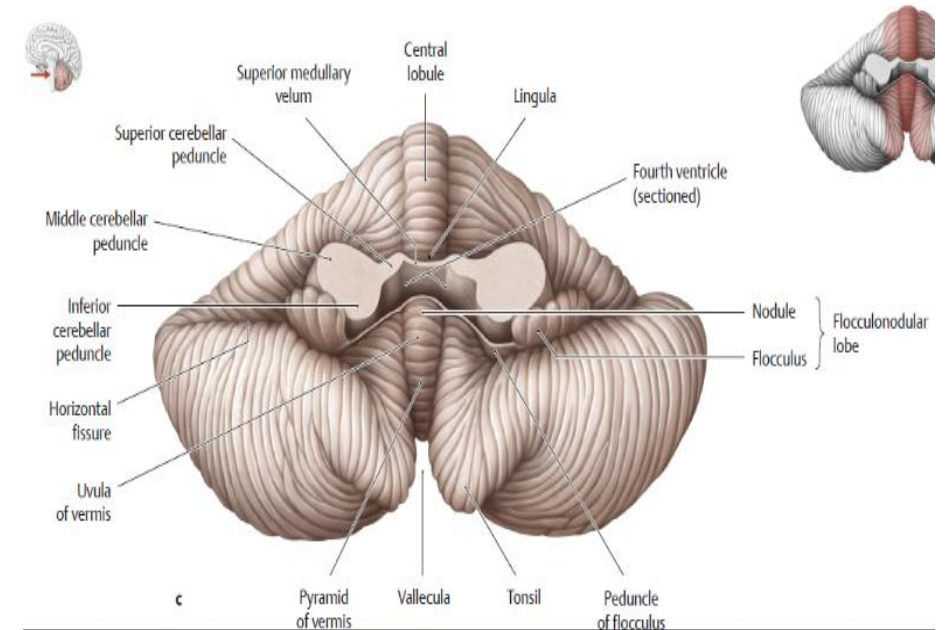
Superior surface



Inferior surface

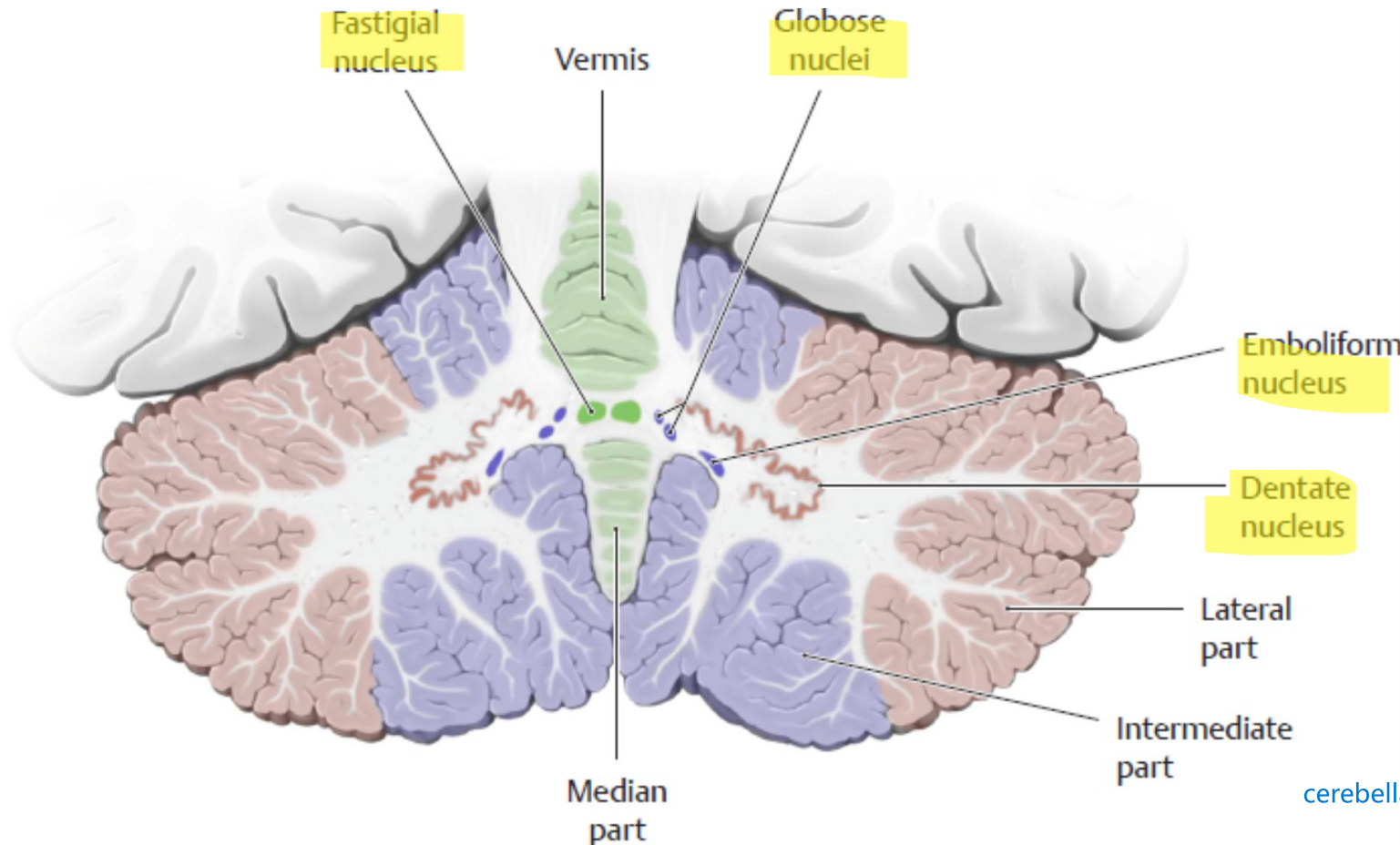


Ventral surface



Cerebellar Anatomy

OBJ. # 2

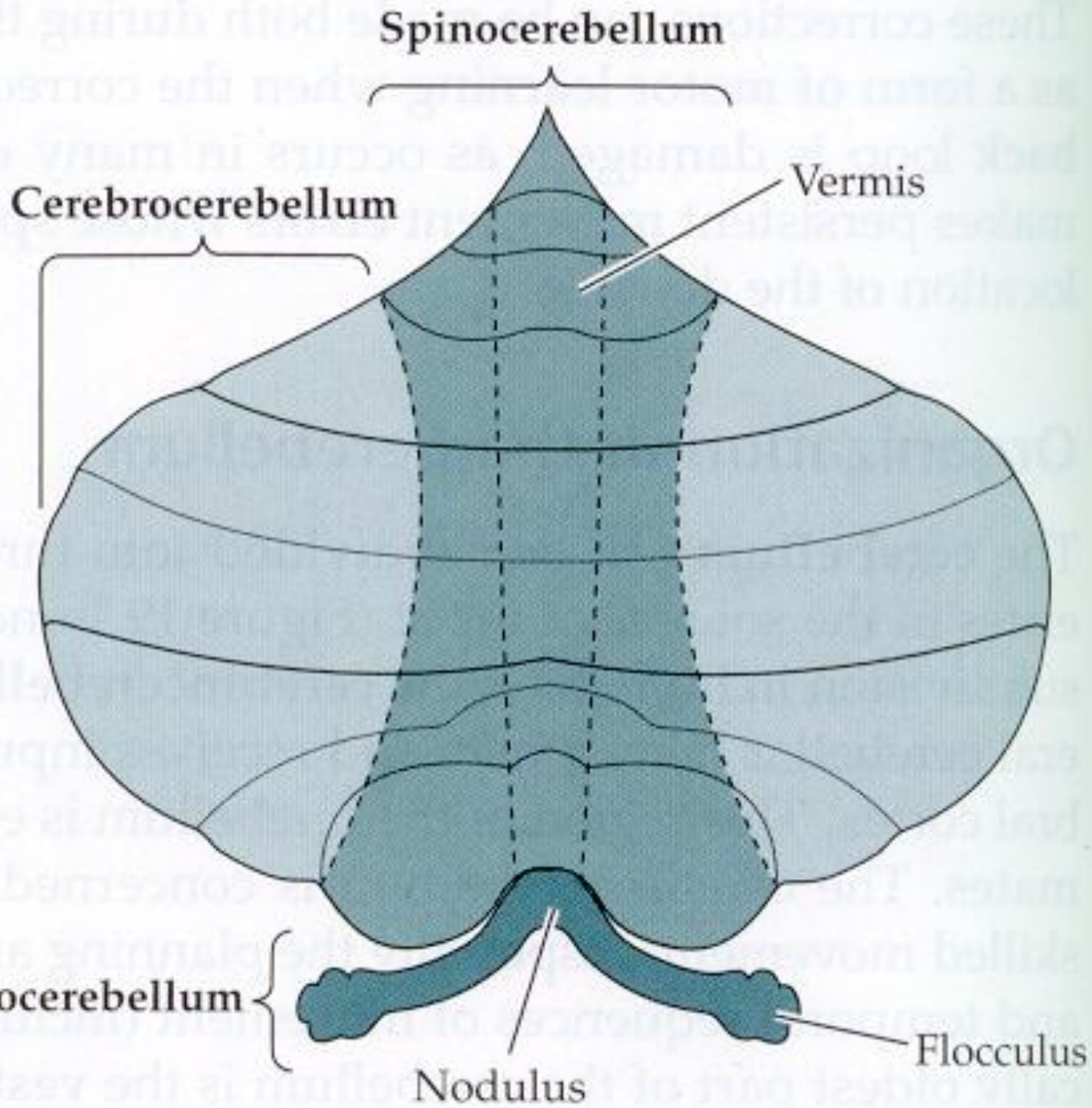


only output pathways

- 4 deep cerebellar nuclei: **dentate, emboliform, globose, and fastigial.**
- **The 4 vestibular nuclei are considered part of the deep cerebellar nuclei because they project directly to the cerebellum**

cerebellar gyri = folia

Don't Eat Greasy Foods (lateral to medial)

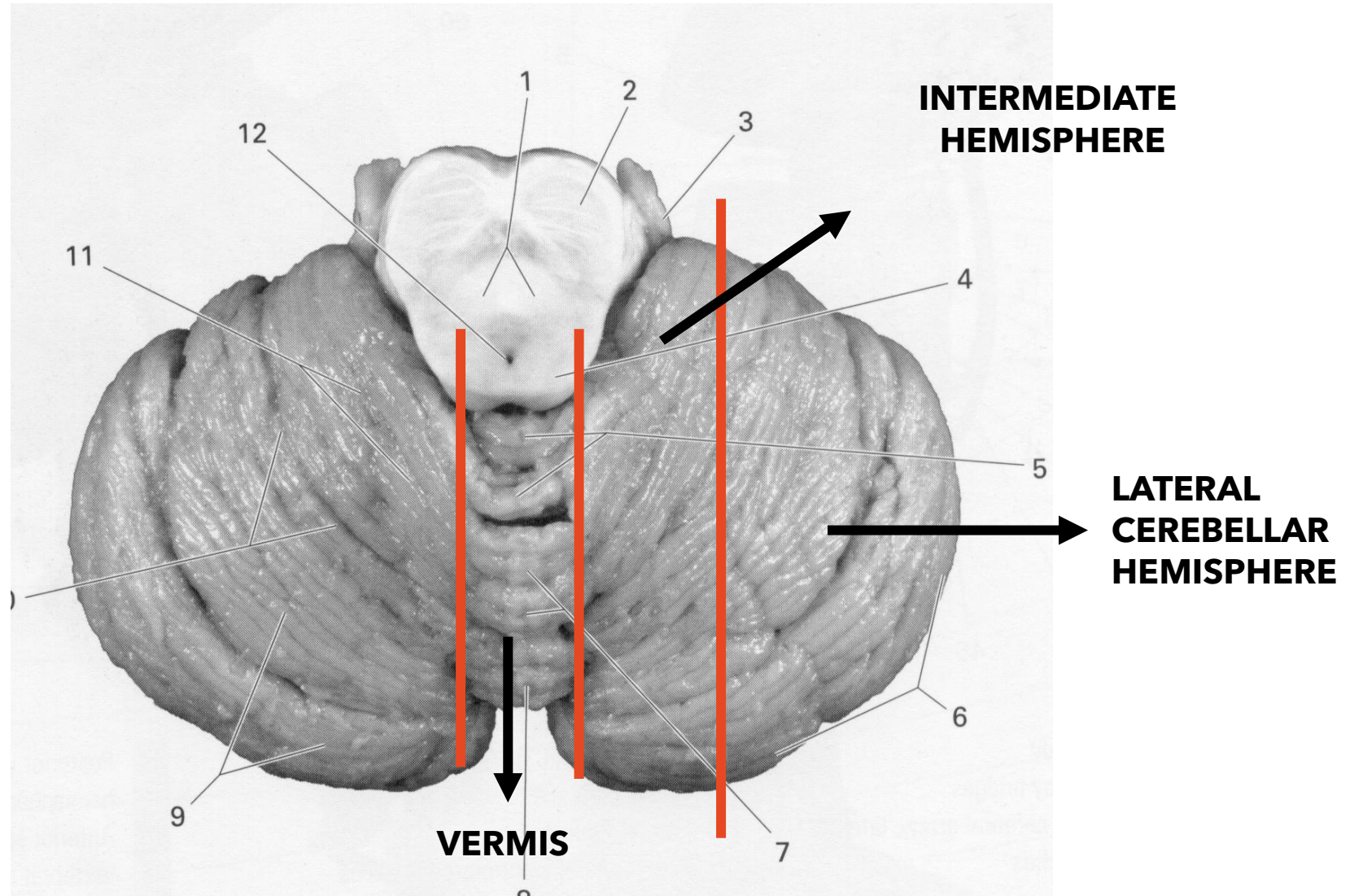


This is an 'unrolled' cerebellum

3 Functional Regions

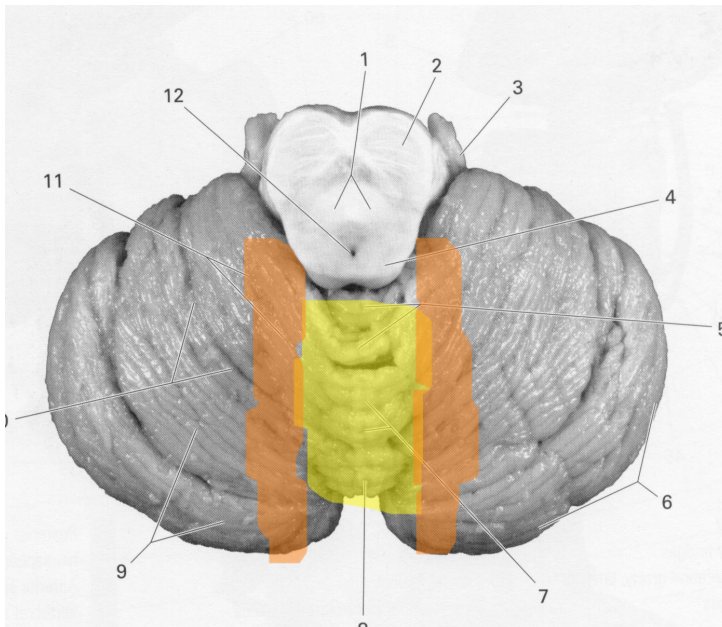
3 Functional Regions

OBJ. # 4



Cerebellar Functional Areas

Spinocerebellum – developmentally older structure:



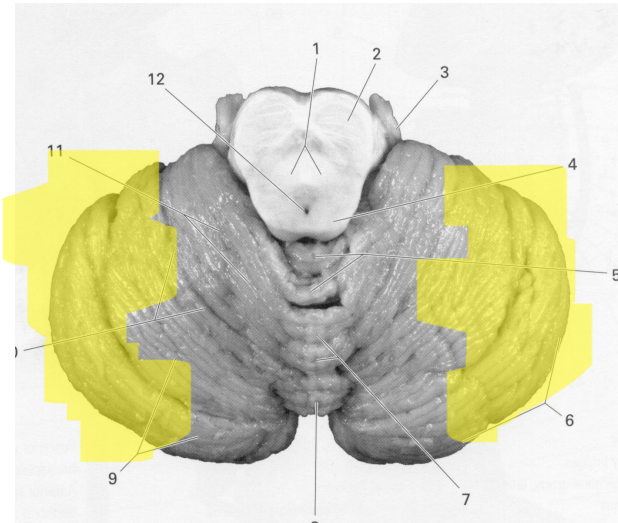
Superior Vermis

**Posture and axial
muscle coordination/
Balance, gait and
muscle tone**

Intermediate
Hemispheres

**Motor coordination of
limb movements**

OBJ. # 4



Cerebellar Functional Areas

Cerebrocerebellum

Lateral Cerebellar
Hemispheres

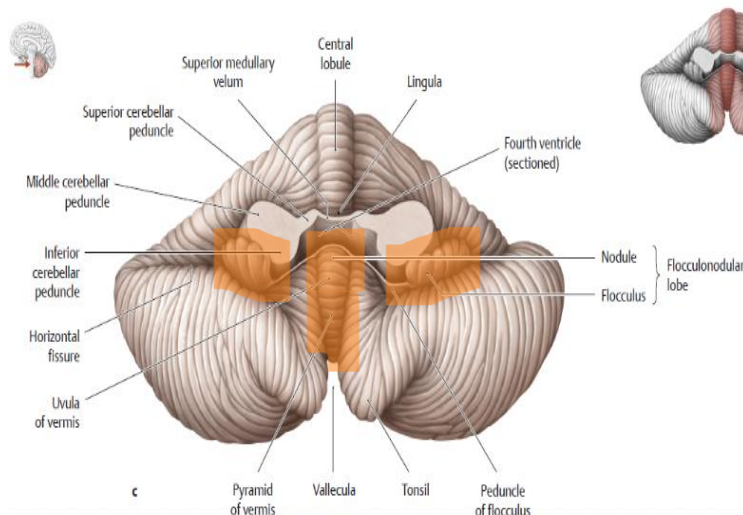
more distal = for limbs

**Motor planning
+ motor learning**

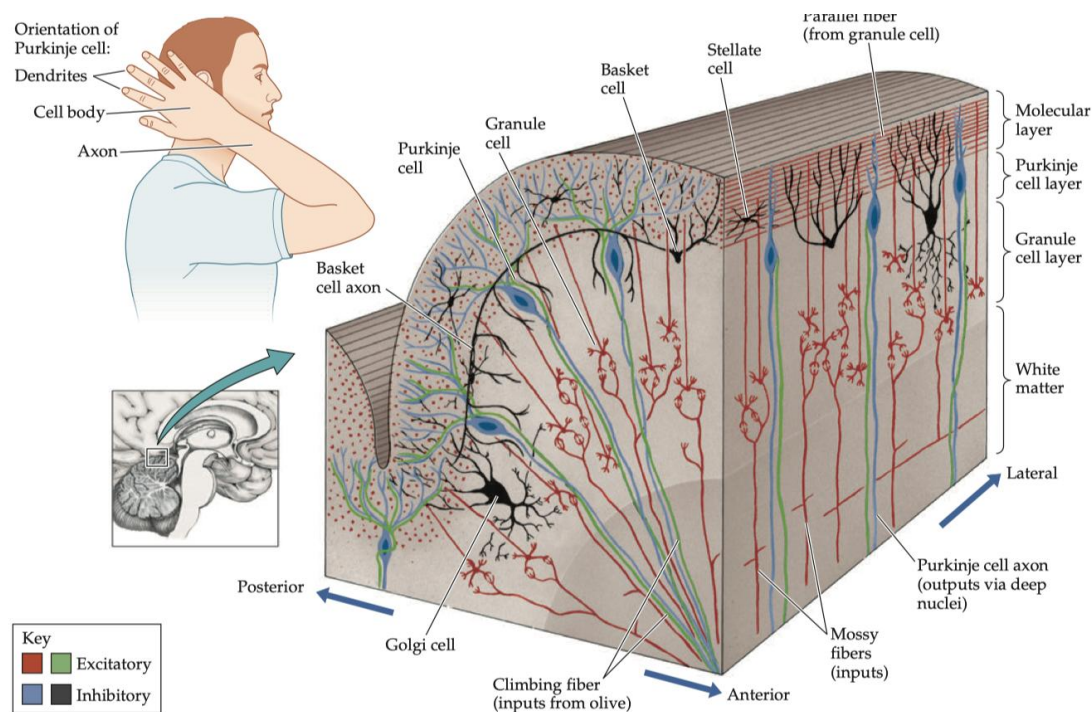
Vestibulocerebellum

Flocculonodular Lobe
+ Inferior Vermis

**Vestibulo-ocular
Coordination
and balance**



Cerebellar Cortex Cellular Organization



3 Layers from outer to inner:

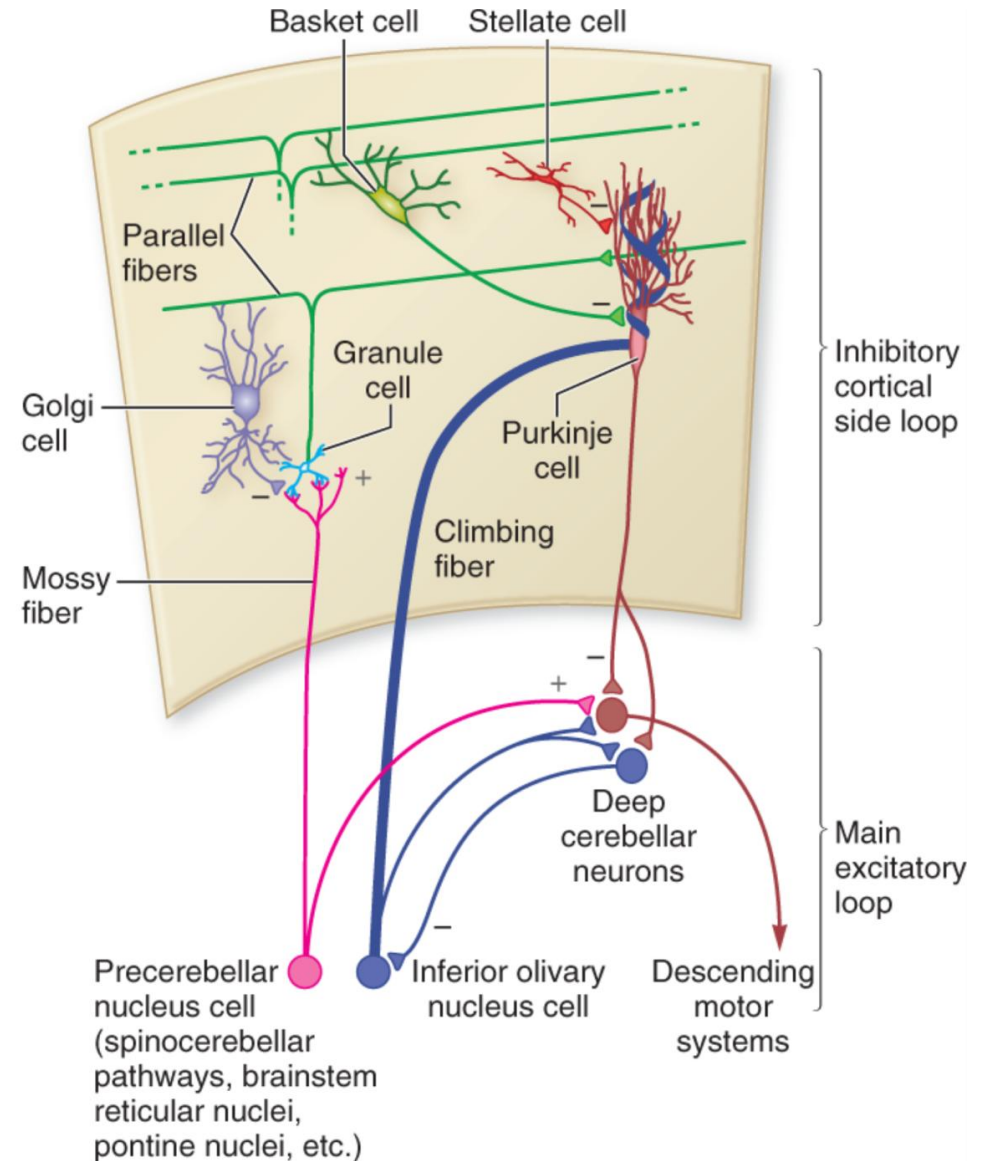
- Molecular
 - Mostly contains granule cell axons
 - Dendrites of purkinje cells
 - Some interneurons
- Purkinje cell
 - Contains the purkinje cells!
 - Second largest cells in the CNS
 - Only axons to leave the cerebellar cortex
 - Purkinje cells use GABA as neurotransmitter -> all output from cerebellar cortex is inhibitor
- Granule cell
 - Densely populated with granule cells and other interneurons, nearly as many cells as the rest of the nervous system
 - Only excitatory neuron in cerebellum (glutamate)

OBJ. # 6

Cerebellar Cortex

Inputs to the cerebellum:

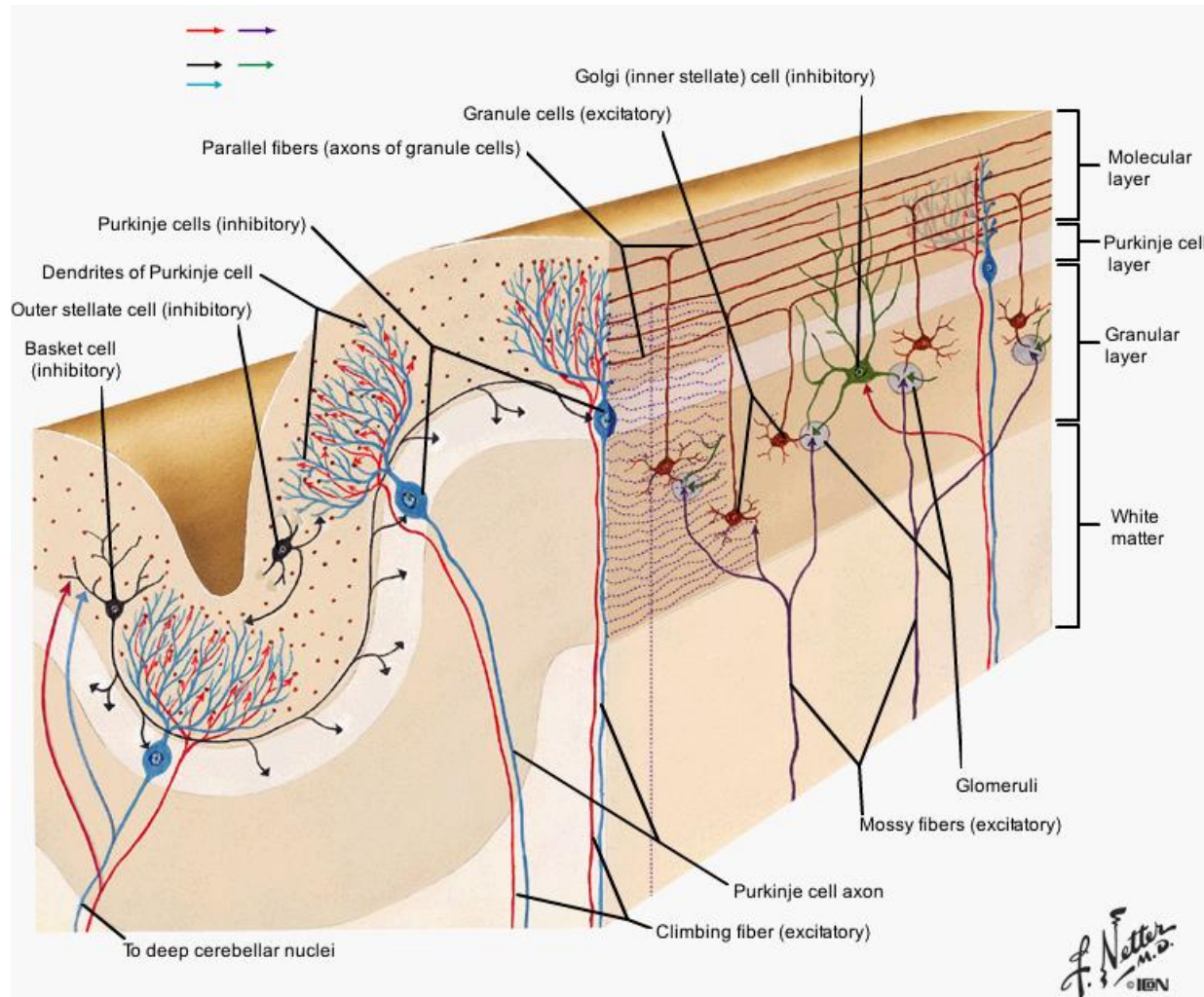
- Climbing fibers
- Mossy fibers



Source: Ropper AH, Samuels MA, Klein JP: *Adams and Victor's Principles of Neurology, Tenth Edition*: www.accessmedicine.com
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Cerebellar Afferents:

OBJ. # 6



Mossy fibers:

- Enter via ICP and MCP
- Coming from pontocerebellar, spinocerebellar, and vestibulocerebellar tracts
- Form excitatory (glutamate) synapses on granule cells

Climbing fibers:

- Enter via ICP
- Projected from olivary nuclei (olivocerebellar tracts)
- Excitatory input directly on purkinje cells
- Important for motor learning

Cerebellar Cortex Projections To Deep Cerebellar Nuclei

only purkinje cells are doing this

Lateral cerebellar hemispheres



Dentate Nucleus

Intermediate Cerebellar Hemispheres



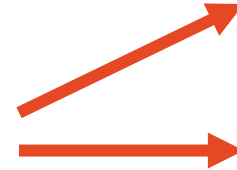
Interposed Nuclei

Superior Vermis



Fastigial Nuclei

Flocculonodular lobe and inferior vermis



Vestibular Nuclei

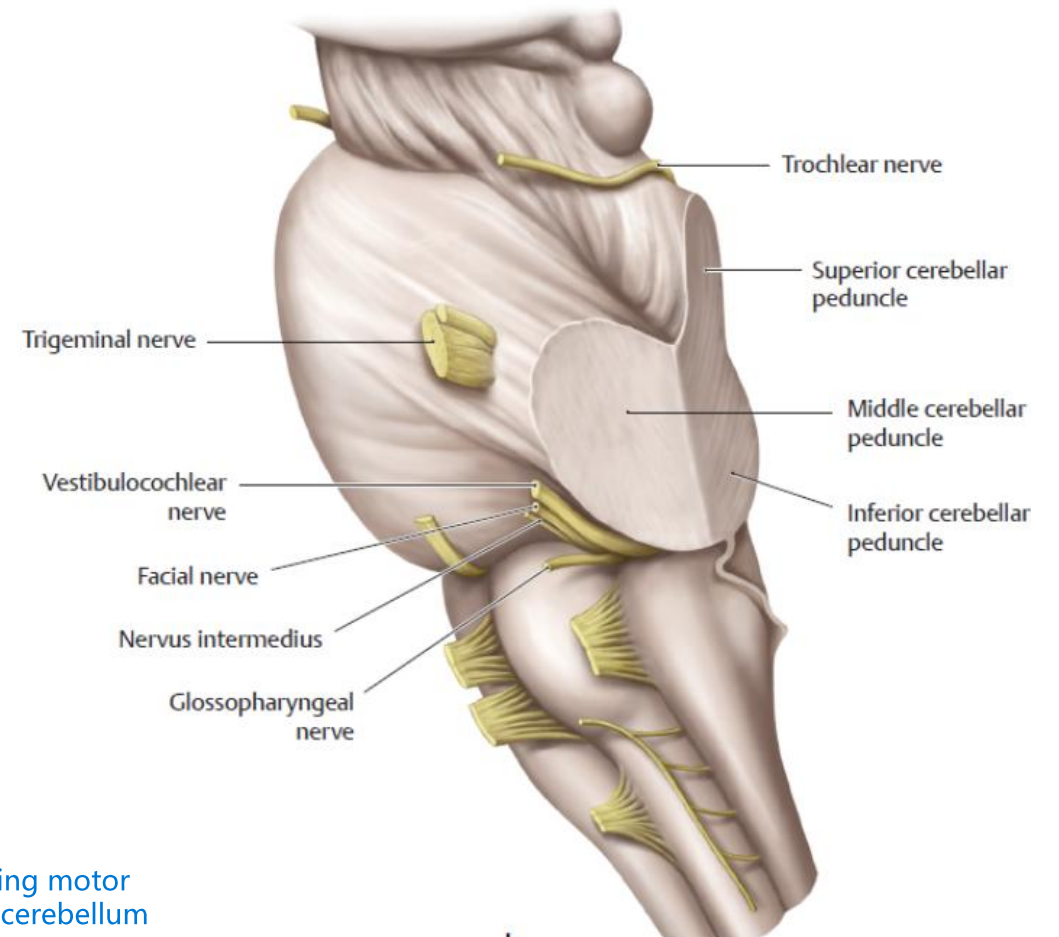
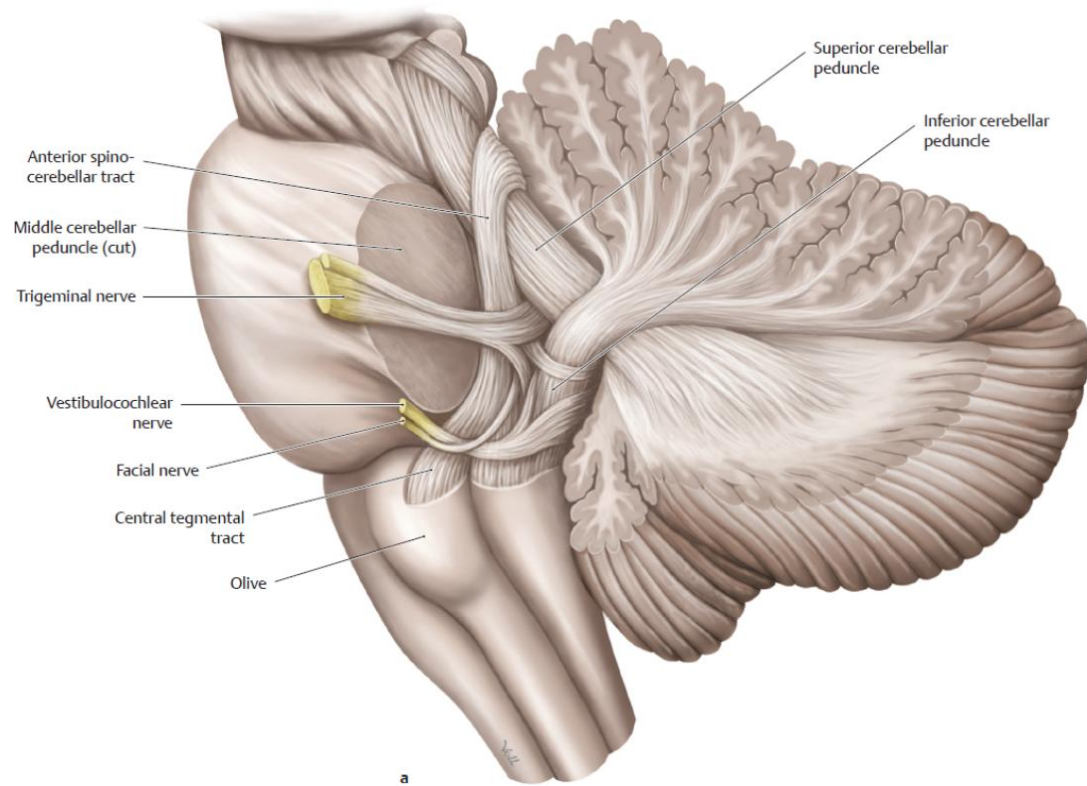
not in cerebellum but part of cerebellar pathways

Cerebellar Cortex Projections To Deep Cerebellar Nuclei

TABLE 15.2 Main Cerebellar Output Pathways

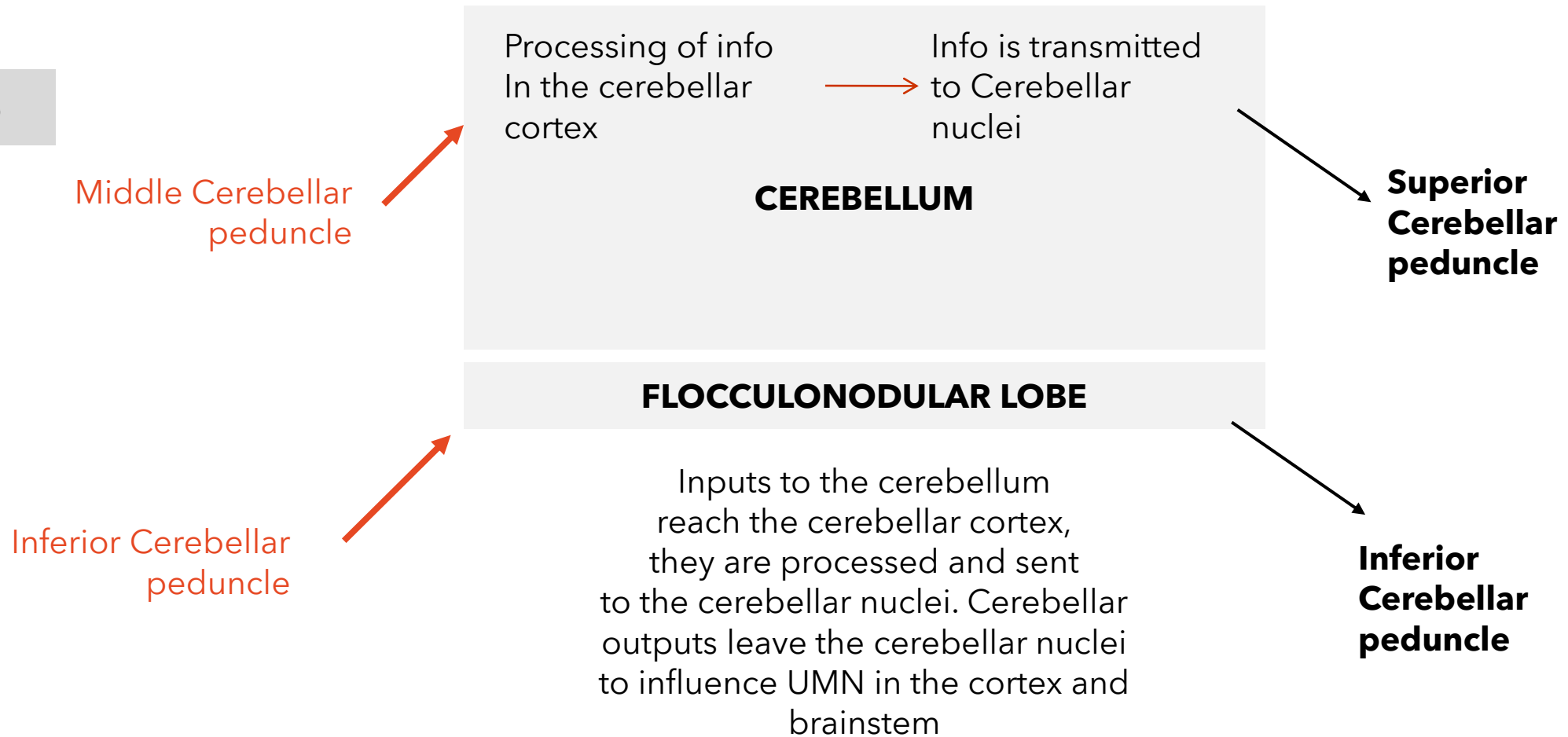
REGION	DEEP NUCLEI	CEREBELLAR PEDUNCLE	MAIN OUTPUT TARGETS OR EQUIVALENT
Lateral hemispheres	Dentate nucleus	Superior cerebellar peduncle	Ventrolateral nucleus of thalamus (VL), parvocellular red nucleus
Intermediate hemispheres	Interposed nuclei	Superior cerebellar peduncle	VL, magnocellular red nucleus
Vermis	Fastigial nuclei	Superior cerebellar peduncle Uncinate fasciculus, ^a juxtarestiform body ^b	VL, tectum Reticular formation, vestibular nuclei
Inferior vermis and flocculonodular lobe	Vestibular nuclei	Juxtarestiform body ^b	Medial longitudinal fasciculus (eye movement pathways)

Cerebellar Peduncles



middle: main descending motor
superior: output from cerebellum

OBJ. # 5



Cerebellar Inputs And Outputs

Cerebellar Inputs

Middle Cerebellar Peduncle

pontocerebellar projecting up into motor cortex

OBJ. # 5

Corticopontine fibers from all brain lobes terminate in pontine nuclei



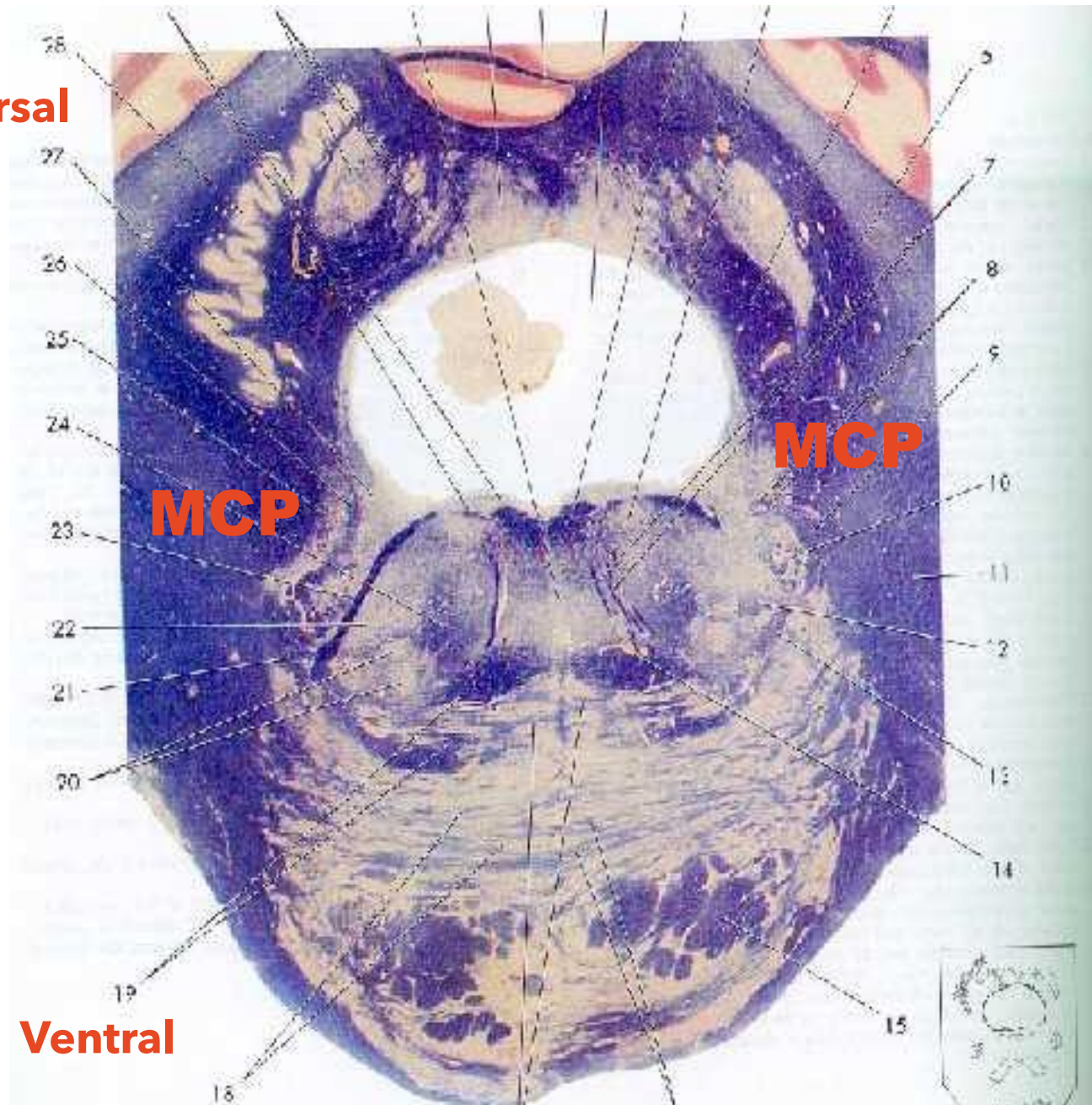
Pontocerebellar fibers cross the midline and form the contralateral middle cerebellar peduncle



They enter the cerebellum as **mossy fibers** to reach the entire cerebellar cortex, except the cortex of the nodulus

OBJ. # 5

Dorsal



MCP

MCP

Ventral



Cerebellar Inputs

Inferior Cerebellar Peduncle

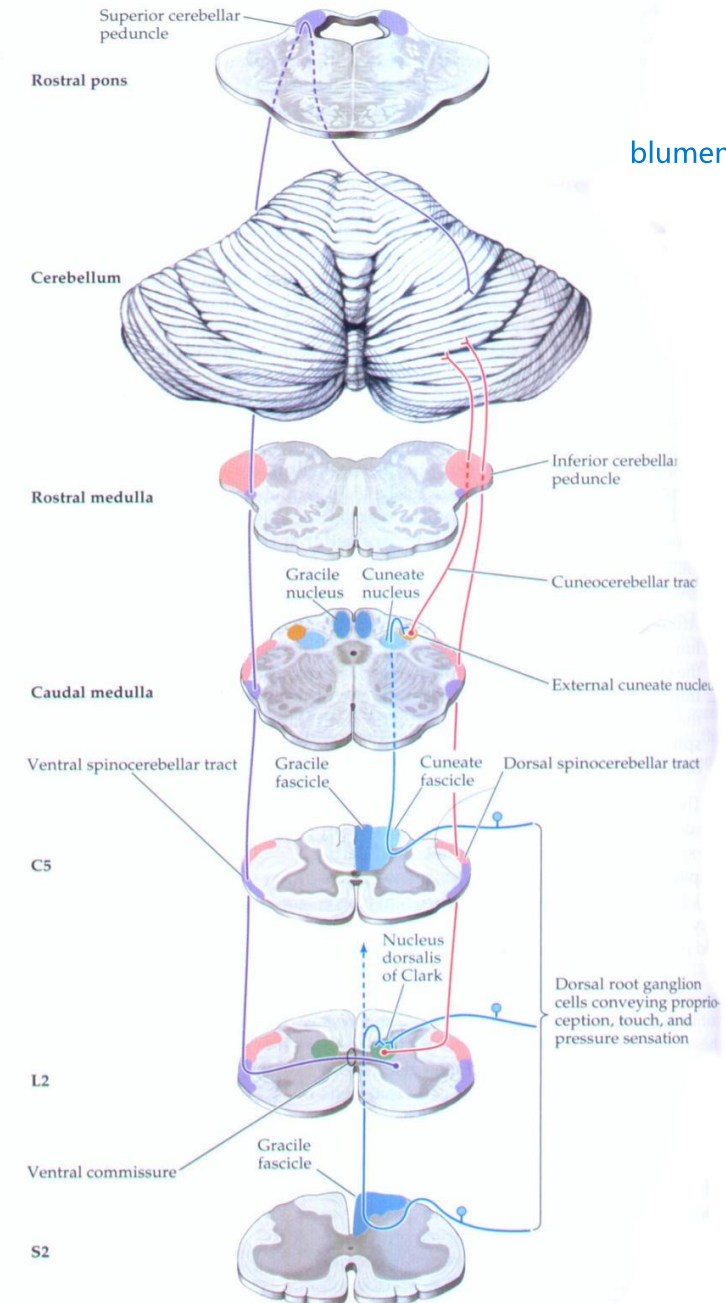
OBJ. # 5

ascending tract

- **Spinocerebellar fibers** travel in several tracts.
 - Dorsal spinocerebellar tract
 - Ventral spinocerebellar tract
 - Cuneocerebellar tract
 - Rostral spinocerebellar (we don't know much about this pathway)
- They convey sensory information about limb proprioception from all levels of the spinal cord. The fibers terminate mostly in the paravermal area and superior vermis as mossy fibers

OBJ. # 5

Spinocerebellar Fibers



blumenfeld text

Cerebellar Inputs

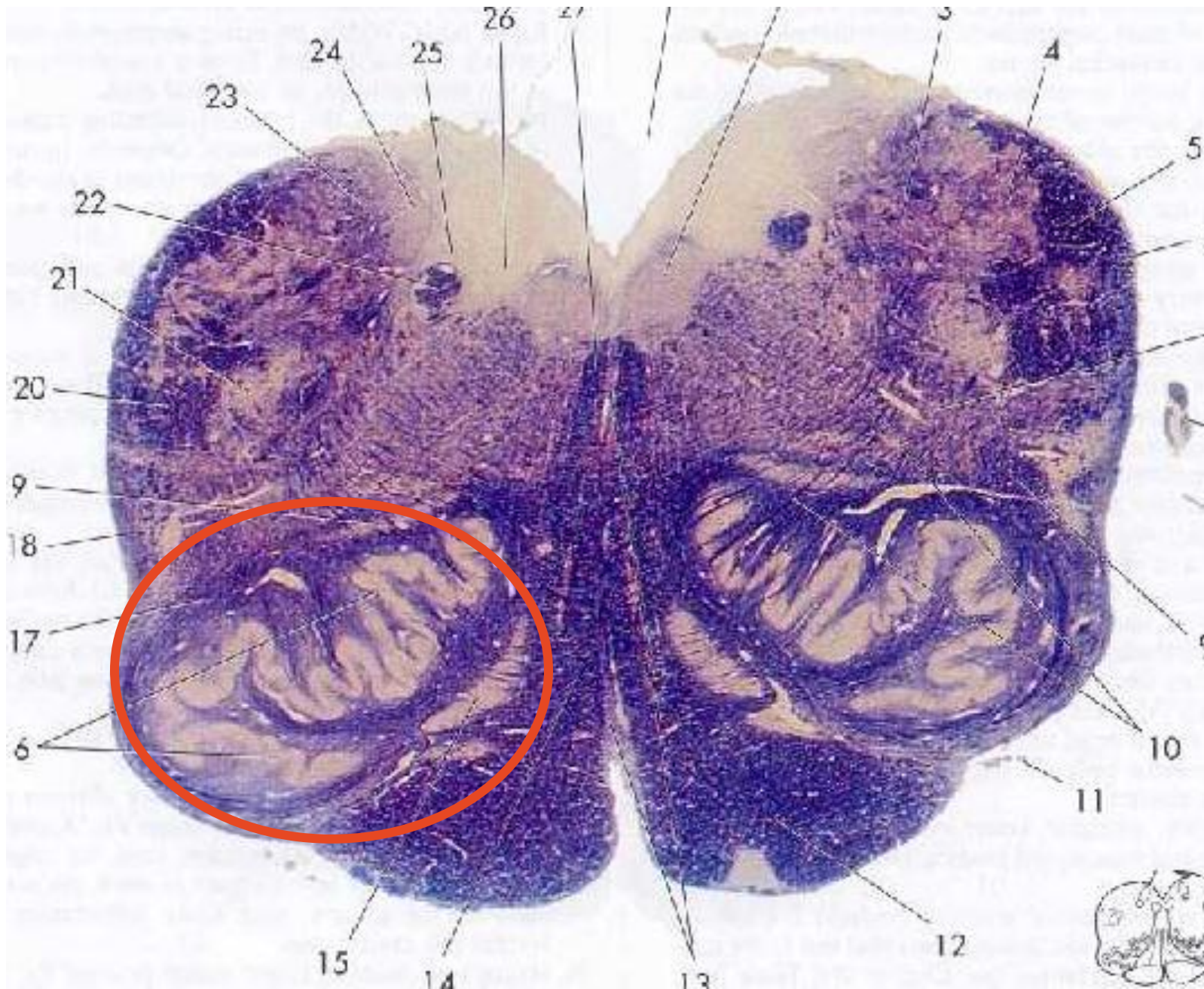
Inferior Cerebellar Peduncle

OBJ. # 5

Spinocerebellar fibers travel in several tracts. They convey sensory information about limb movement from all levels of the spinal cord. The fibers terminate mostly in the intermediate zone and superior vermis as mossy fibers

Olivocerebellar fibers cross the midline to enter the contralateral cerebellum as **climbing fibers** to terminate in the entire cerebellar cortex

OBJ. # 5



**Inferior
Olivary
Nucleus**

Cerebellar Inputs

Inferior Cerebellar Peduncle

OBJ. # 5

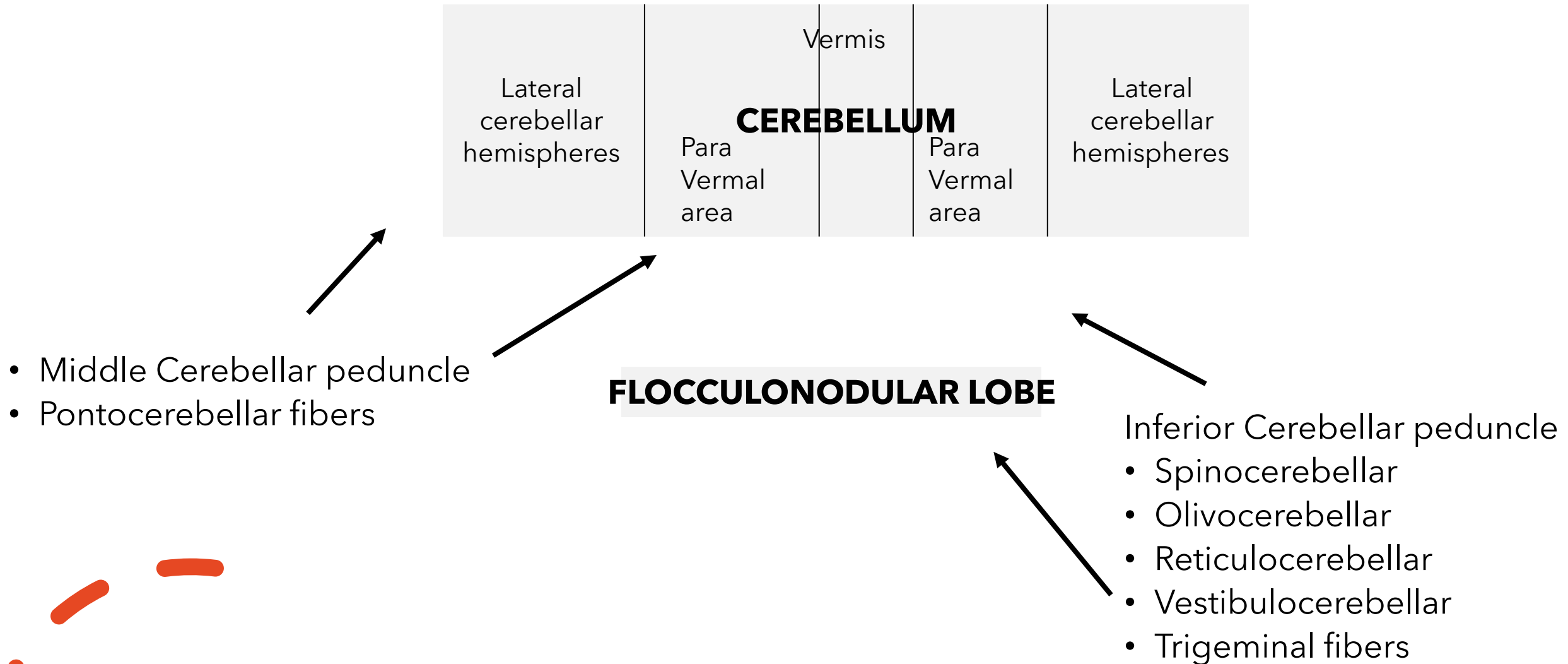
Spinocerebellar fibers travel in several tracts. They convey sensory information about limb movement from all levels of the spinal cord. The fibers terminate mostly in the intermediate zone and superior vermis as mossy fibers

Olivocerebellar fibers cross the midline to enter the contralateral cerebellum as climbing fibers to terminate in the entire cerebellar cortex

Primary and secondary vestibular fibers project via the juxtarestiform body, a subpart of the inferior cerebellar peduncle, as **mossy fibers** to terminate in the flocculonodular lobe and inferior vermis

Cerebellar Inputs

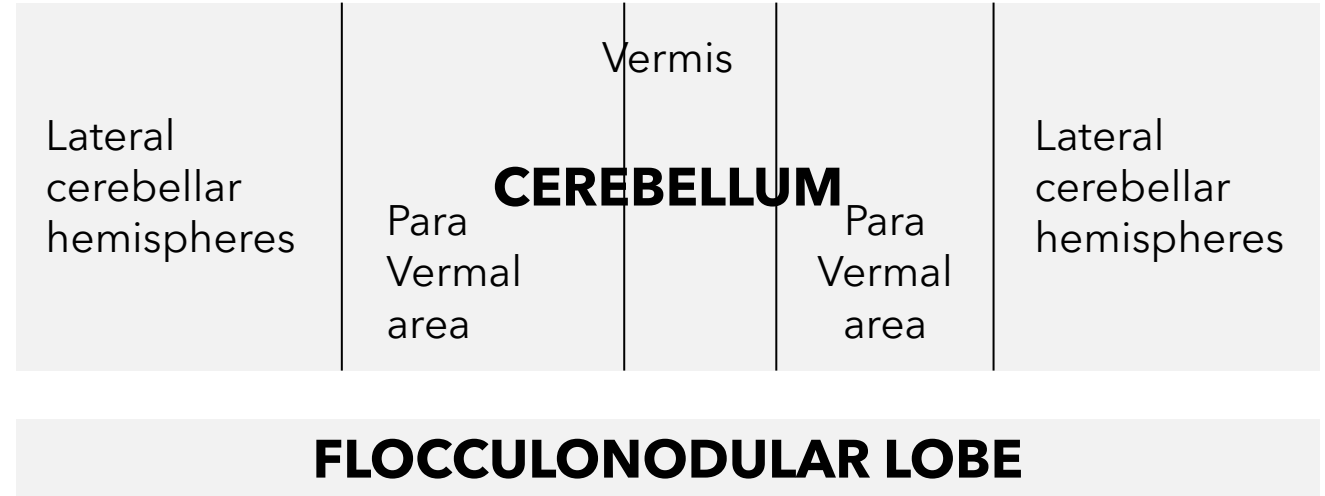
OBJ. # 5



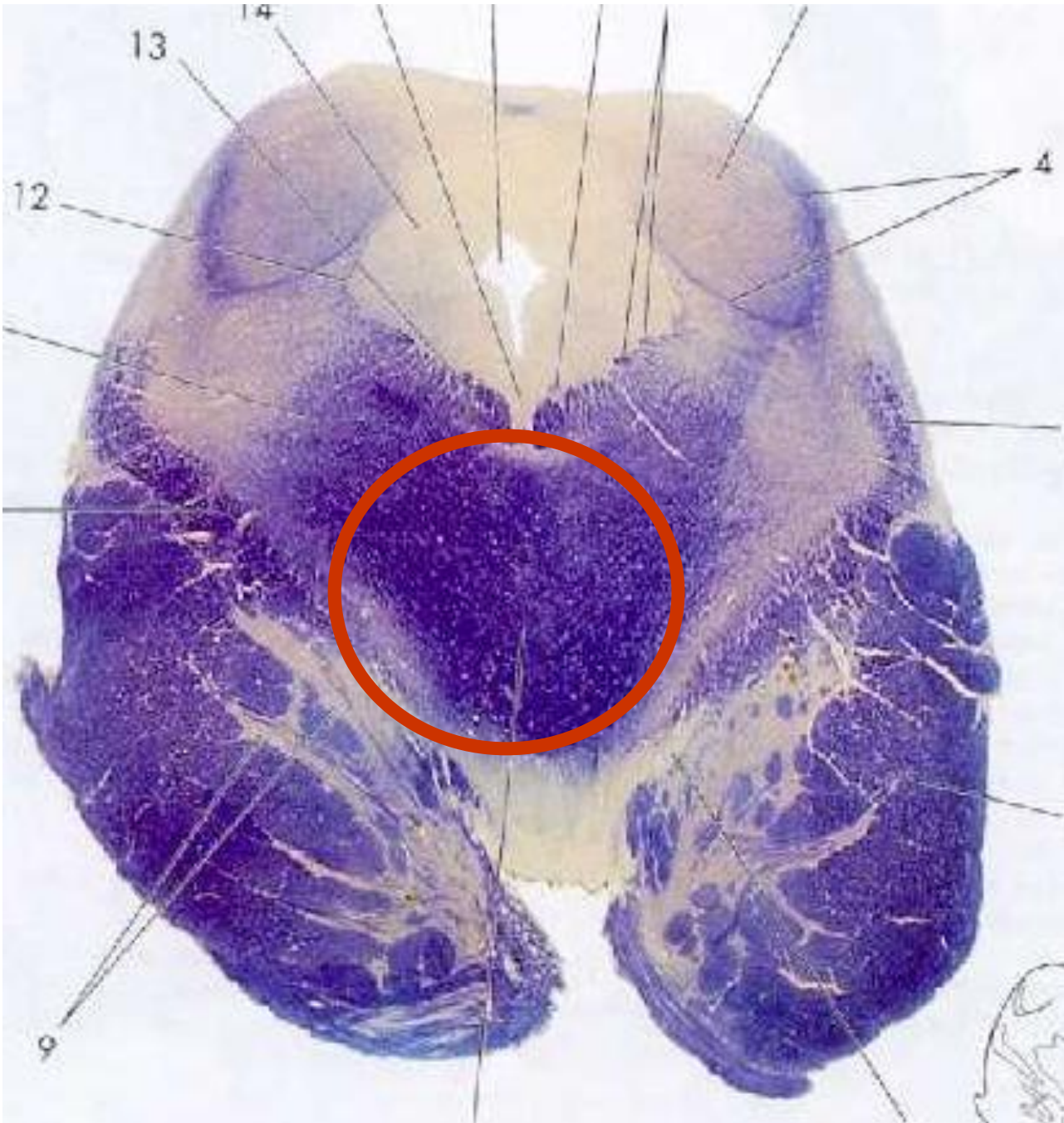
CEREBELLAR OUTPUTS

OBJ. # 7

- **Superior Cerebellar peduncle - decussates in the caudal midbrain**
 - Thalamus VA/VL ----- Motor cortex (contralateral)
 - To influence the corticospinal tracts
 - Red Nucleus (contralateral)



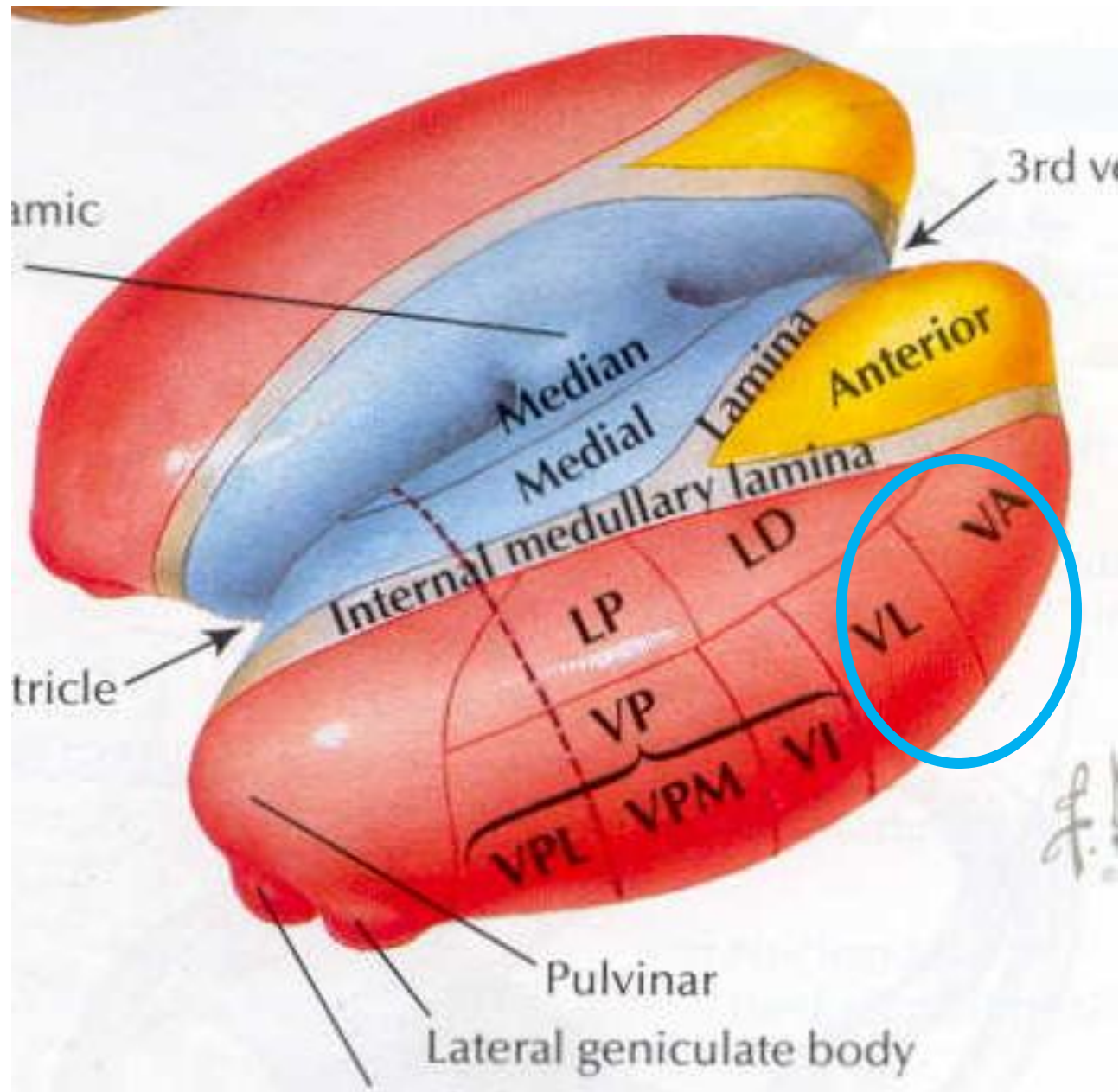
- **Inferior Cerebellar peduncle** (ipsilateral)
 - To reticular formation
 - To influence reticulo-spinal tracts
 - **To vestibular nuclei**
 - To influence vestibulo-spinal tracts and gaze centers



Superior Cerebellar Peduncles decussation

OBJ. # 7

OBJ. # 7

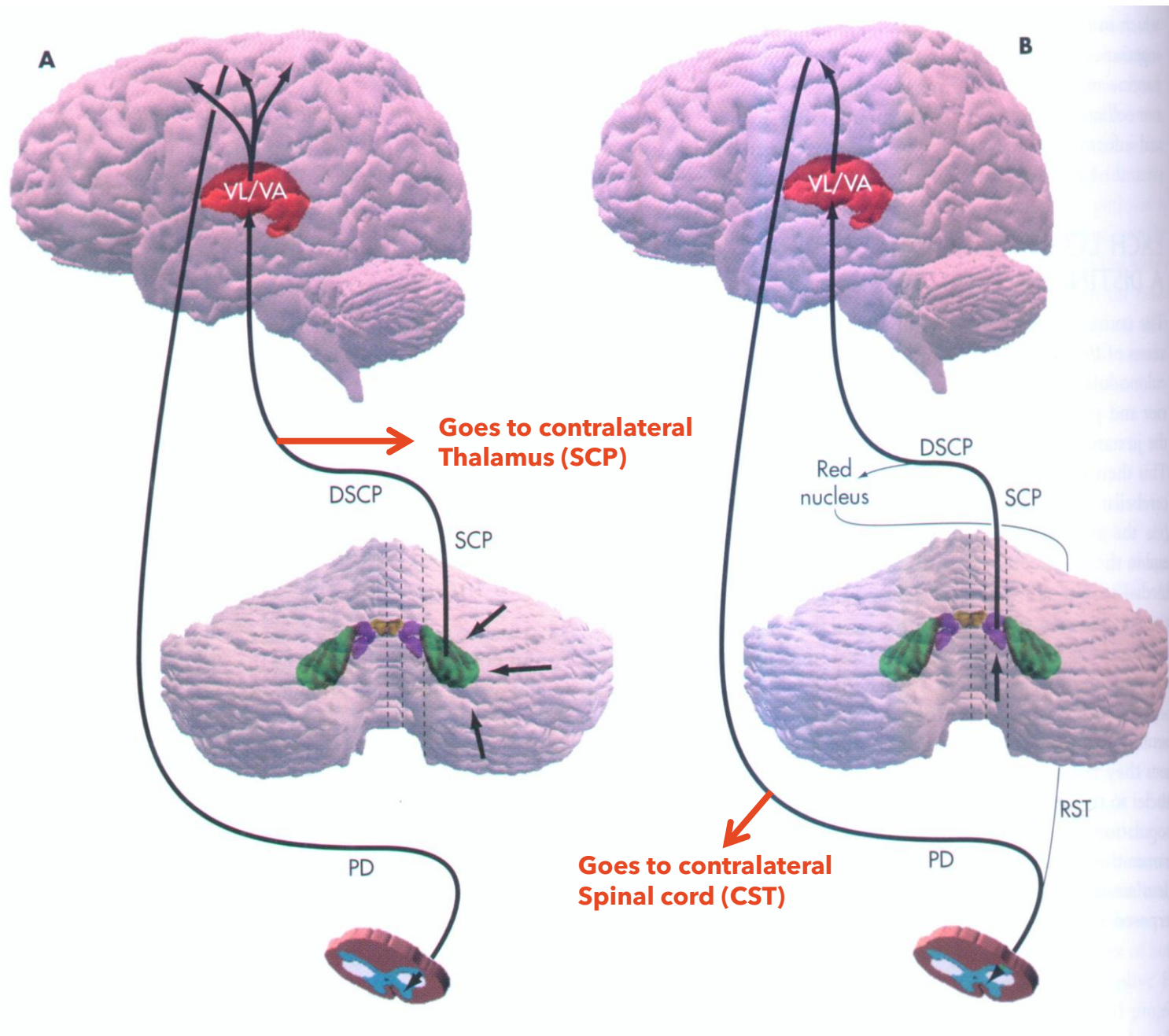


major projections from contralateral sup cerebellar peduncles

Thalamus



Clinical Correlates



Why do cerebellar lesions produce ipsilateral deficits?

OBJ. # 10



Cerebellar Lesions

Lack of motor coordination due to cerebellar damage is called ATAXIA

OBJ. # 8

global term

ATAXIA =

Inability to perform smooth, coordinated movements that result in irregularity and fragmentation of the normal motor sequence

Lesions to the cerebellum or cerebellar pathways produce different type of abnormalities:

- Incoordination or ataxia
- Intention tremor
- Disorders of equilibrium and gait
- Decreased muscle tone
- Scanning dysarthria
- Stability of eye movements is affected – nystagmus
- Frequently nausea and vomiting are present



Ataxic Movements

Appendicular ataxia

OBJ. # 8

poor predictions of distance

Dysmetria / past pointing

- Finger-nose-finger test
- Heel-shin test



Wavering movement
with over- or
undershooting

Dysrhythmia

- Precision finger tapping
- Tapping on the floor



Abnormal timing of
movement

Dysdiadochokinesia /
Adiadochokinesia



Abnormal coordination
of Rapid alternating
movements



Ataxic Movements

OBJ. # 8

Gait is Wide-based,
unsteady with tendency to
fall towards lesioned side

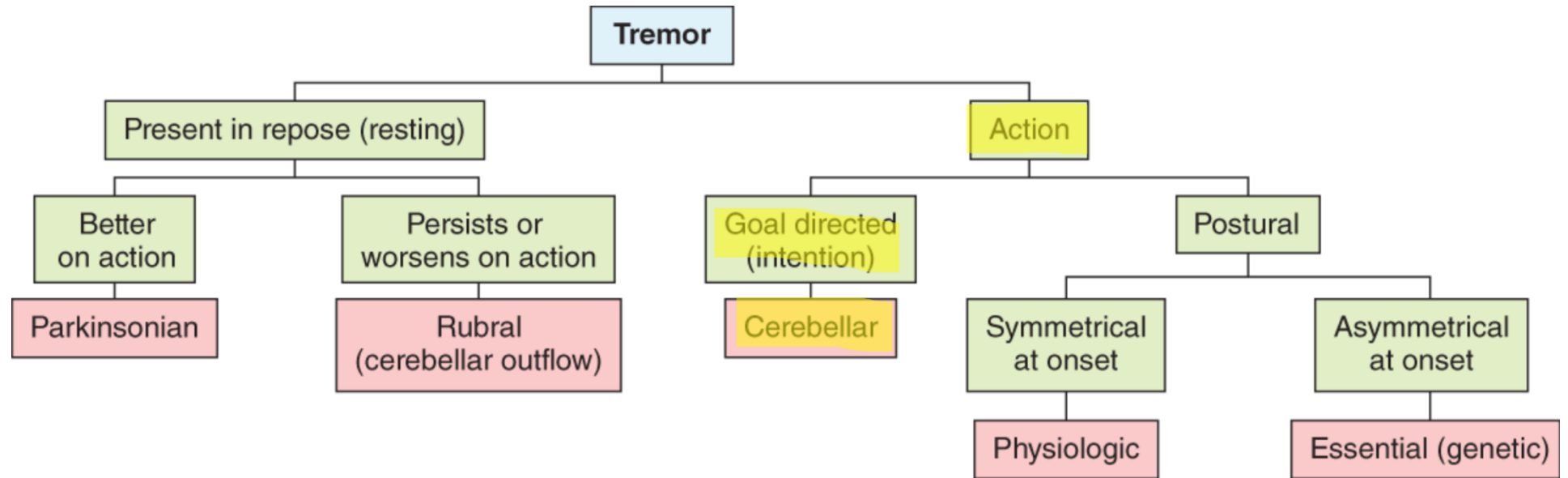
- Make the patient walk → Observe the gait
- Tandem gait impaired → Patient will fall or deviate towards one side
- Romberg → Patient is unable to maintain position with eyes open or closed. This is technically a negative Romberg test.

Differential Diagnosis of Tremor

OBJ. # 9

There are basically 2 types of tremors:

- Action tremor
- Repose (resting) tremor

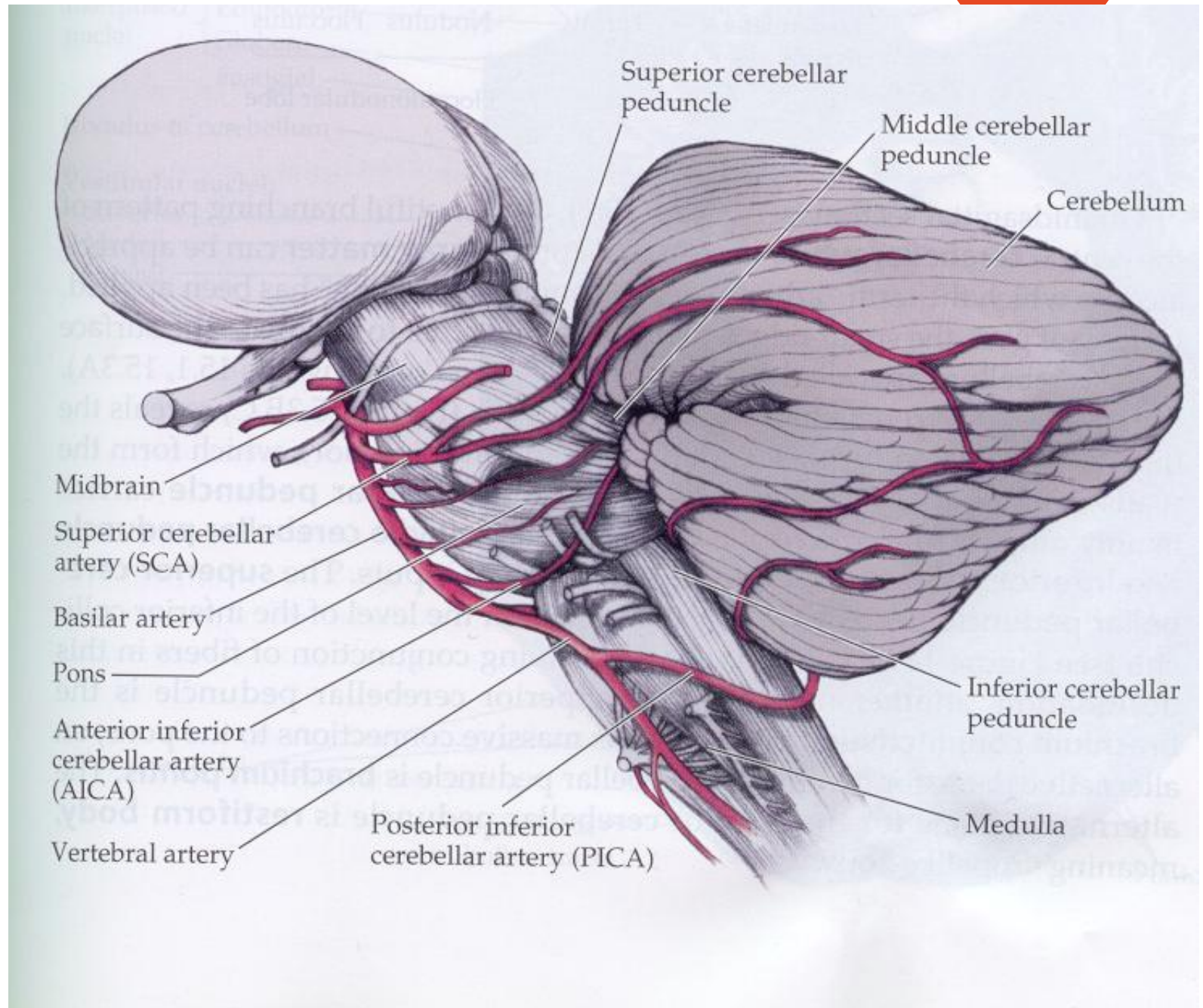


Source: Ropper AH, Samuels MA, Klein JP: *Adams and Victor's Principles of Neurology*, Tenth Edition: www.accessmedicine.com
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From: Part 2. Cardinal Manifestations of Neurologic Disease
Adams and Victor's Principles of Neurology, 10e, 2014

Vascular Supply

OBJ. # 10



Cerebellar Infarcts

OBJ. # 10

3 main arteries that supply the cerebellum:

- Superior cerebellar artery
- Anterior inferior cerebellar artery (AICA)
- Posterior inferior cerebellar artery (PICA)

basilar branches

SCA branch

* Brainstem supply is variable for each of these arteries as some brainstem structures receive more supply from basilar perforating branches

Primary symptoms include:

- Vertigo
- Dizziness
- Nausea and Vomiting
- Gait unsteadiness
- Limb clumsiness
- Headache
- Dysarthria
- Diplopia

Vascular Supply

OBJ. # 10

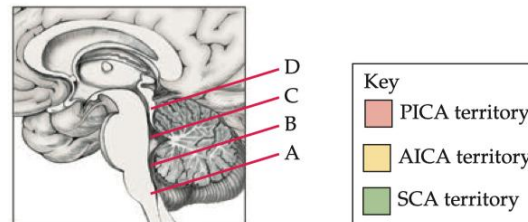
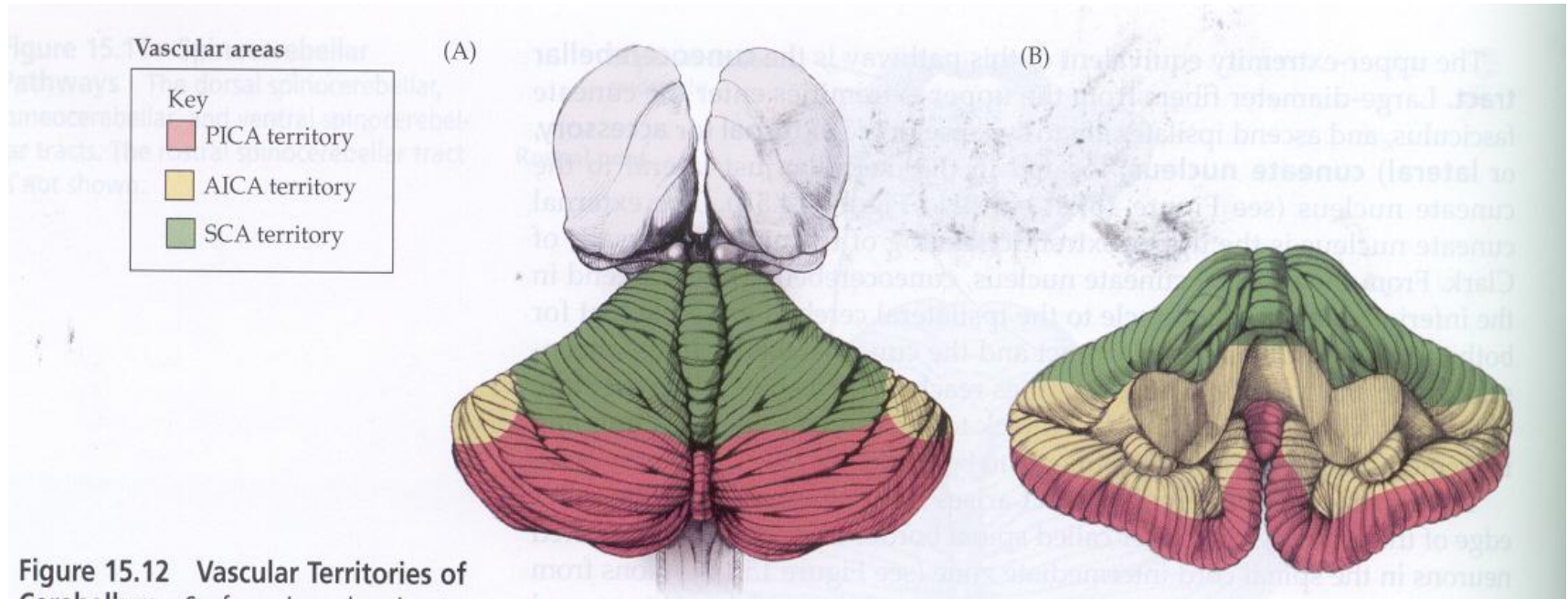
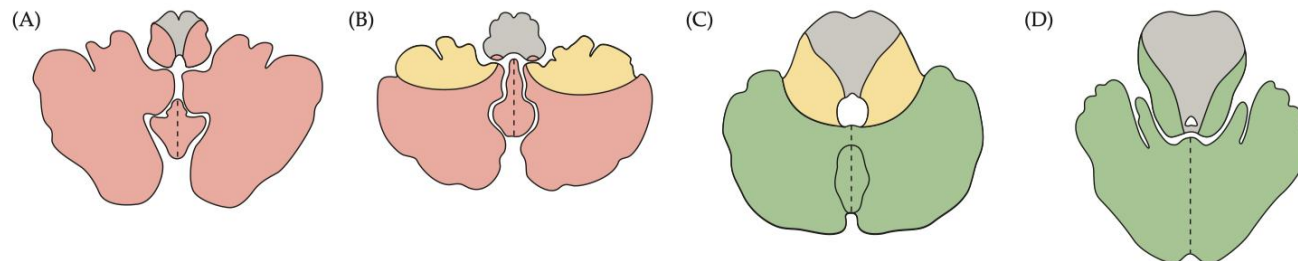


FIGURE 15.13 Vascular Territories of the Cerebellar Arteries Reviewed in Axial Sections (A) Caudal cerebellum and mid-medulla. (B) Caudal cerebellum and rostral medulla. (C) Mid-cerebellum and mid-pons. (D) Rostral pons and rostral cerebellum. PICA, posterior inferior cerebellar artery; AICA, anterior inferior cerebellar artery; SCA, superior cerebellar artery.



Cerebellar Infarcts

OBJ. # 10

It is difficult to localize cerebellar strokes between the three cerebellar arteries:

- They provide variable blood supply to the cerebellum
- Brainstem involvement is variable due to basilar artery perforating branches

ipsi bc double crossing

Symptoms that can be common to SCA, AICA, PICA syndromes:

- Ipsilateral ataxia
- Dysarthria
- Ipsilateral Horner's syndrome
- Ipsilateral facial pain/temp loss
- Contralateral body pain/temp loss

Symptoms suggestive of AICA:

- Ipsilateral hearing loss (CN VIII)
- Ipsilateral facial paralysis

Symptoms suggestive of SCA:

- Contralateral hearing loss (crossing fibers of lateral lemniscus)
- Contralateral 4th nerve palsy
- Relative lack of vertigo/nausea

Cerebellar Disorders Etiology

OBJ. # 11

Acute Ataxia

- Toxic
- Ischemic stroke
- Hemorrhagic stroke
- Traumatic hematoma
- Multiple Sclerosis
- Infections / brain abscess
- Para-neoplastic syndrome

Chronic Ataxia

- Cerebellar metastasis: lung and breast carcinomas, melanomas
- Multiple sclerosis
- Chronic ingestion of toxics
- Degenerative disorders
- Genetic ataxias

alcohol causes a toxic ataxia - purkinje cells sensitive to ethanol

input, output, decussation, mossy vs climbing fibers
purkinje is only output
deep cerebellar nuclei is output to body